

Lily Memo

AI-Powered Study Companion for STEM Education

DSH Hacks V1 · AI × STEM Education · github.com/sikunxchan/lily_memo

The Problem

STEM students spend most of their study time on passive input - reading textbooks, copying notes, watching lectures. Cognitive science consistently shows that active recall and spaced repetition are far more effective for retention, yet turning your own material into practice problems is slow and difficult, and off-the-shelf problem sets never match what you actually studied.

What It Does

Lily Memo turns any study material - typed notes, PDFs, textbook photos, live lecture audio - into active learning. The built-in AI tutor Lily reads it and generates explanations, diagrams, quizzes, and full exam-style problem sets in real time.

AI Tutor "Lily"	Chats over your notes, PDFs and images. Generates Mermaid diagrams, charts, geometry figures and 6 quiz types inserted directly into notes. Accuracy-first persona: corrects mistakes instead of agreeing.
Floating "Sikun"	A draggable on-screen AI character for instant answers: note summaries, to-do extraction, PDF reading, translation, and writing annotations onto PDF pages.
Practice Problem Sets	Multiple choice, written, fill-in, true/false, reading and chart questions - solved full-screen with scoring, explanations, attempt history and a "brutal" difficulty tier.
PDF -> Markdown	Vision-based transcription of any PDF (scanned and handwritten included), saved as a note or .md file. Math is rendered as LaTeX.
Live Lecture Summary	Record class audio -> auto-generates Cornell-style notes, key terms and 10 practice questions.
Rich Notes	LaTeX math, code blocks, tables, handwriting blocks, sketches, wiki-style note links and a graph view of connections. Fully offline (IndexedDB).
Gamified Tracking	Pomodoro sessions, level system (Lv 1-500), 35+ badges, trophy room, streaks, weekly schedule and tasks.
Live Sync	Optional cross-device sync of everything via a shared key - conflict-safe with atomic compare-and-set.

How It Was Built

Next.js 16 / React 19 / TypeScript - installable offline-capable PWA · Gemini 2.5 Pro / Flash / Flash-Lite with 3-tier model fallback, extended thinking and vision input · TipTap editor with custom nodes (Mermaid, Chart.js, geometry, Q&A cards, handwriting) · Dexie (IndexedDB) local-first storage · Upstash Redis live sync with a Lua compare-and-set script preventing concurrent-write data loss · pdf.js, KaTeX, Mermaid, Chart.js, Web Audio API.

Voice features use MediaRecorder + WAV re-encoding instead of the Web Speech API, so they work inside installed PWAs on iOS where Web Speech silently fails.

Impact

Any student with a browser can use Lily Memo for free in English with zero setup (server-side API key). It targets STEM learners who have material but no time to turn it into practice: a lecture recording becomes structured notes in minutes, and a PDF chapter becomes a 20-question problem set with a single prompt.

Turn passive study material into problems you can solve.

Built with Next.js · Gemini 2.5 · Dexie · Upstash Redis · TipTap